Leading Causes of deaths in United States of America from 1999 to 2017

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Abstract—This Paper employs a novel 3D world visualisation, augmented by a volcano heatmap and point layer overlay, to examine the primary causes of mortality in the United States from 1999 to 2017. This method offers a dynamic and richly spatially-depicted view of national mortality patterns. The volcano heatmap illustrates regions with higher death rates by projecting mortality data onto a three-dimensional globe. This allows for the visual representation of the intensity and geographic distribution of different causes of death across time. This interactive visualisation highlights areas with serious health issues in addition to showing changes in mortality trends. The analysis's conclusions have significant significance for public health policies. In the end, our research seeks to assist policymakers and public health professionals in addressing health disparities.

Index Terms—3D Globe Visualization , Volcanic Heatmap, Point layers , Mortality Trends , Leading Causes of Death , Public Health Analysis , Geographic Health Disparities , Temporal Mortality Patterns , Spatial Data Representation , Health Intervention Strategies , Data-Driven Public Health , US Mortality Data (1999-2017) , Preventive Health Measures

1 Introduction

Understanding the main causes of death is essential for improving public health and developing effective strategies to prevent disease. This Paper examines the leading causes of death in the United States from 1999 to 2017, using a unique visualization method to make the data easier to understand and analyze.

Traditionally, mortality data has been shown through basic charts and tables, which can make it difficult to see patterns and trends over time and across different regions. The initial step focused on exploring various sources to identify effective ways to visualize the data. Using the Five Design-Sheet (FDS) approach, several design possibilities were considered. The first sheet included a range of ideas such as representing data with bar charts, mapping it onto 2D state maps, or using a 3D globe for spatial visualization. Other concepts included heatmaps, candlestick charts, and point-based plots. In the second stage of the Five Design-Sheet (FDS) process, a 2D map chart was created. This design included interactive elements such as year-specific buttons and disease filters. Additionally, hovering over specific areas on the map revealed the corresponding disease name on the screen. Moving to the third stage, the design transitioned to a 3D globe visualization that used candlestick bars and incorporated the same filters for years and diseases. Another approach considered at this stage was a heatmap globe, which provided an efficient way to visualize data while allowing filters for years, diseases, and states to be applied. These early design ideas laid the groundwork for selecting the most effective techniques to present the data clearly and meaningfully.

The 3D globe was chosen because it presents data in a more intuitive and interactive way compared to traditional methods. The volcanic heatmap improves clarity by using color intensity to depict regions with higher mortality rates, resembling the visualization of heat on a volcanic map. This method effectively showcases how mortality patterns vary over time and across different regions for various causes of death. Incorporating a point layer further enhances the visualization by pinpointing specific locations, stats or data points of interest. This addition provides detailed spatial context, enabling a more precise analysis of localized patterns while retaining the broader trends displayed by the heatmap. Together, these features present a detailed and comprehensive perspective on mortality trends nationwide.

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Research Question:

1. Which visualization methods most effectively highlight significant trends and patterns in mortality data over time, and how can these be implemented in interactive 2D and 3D formats?
2. What visualisation strategies can effectively communicate actionable insights for reducing unnecessary deaths and enhancing public health, based on historical and forecasted data?

Aims and Objectives:
By using this advanced visualization technique,

1. this paper aims to reveal patterns in mortality data that might not be obvious with standard methods.
2. to conduct a critical self-evaluation of the proposed visualization tool in terms of its overall strengths and limitations.

2 Background / Literature Review

2.1 Disparities in Kidney Cancer Mortality Among Hispanic Populations in the US:

Exploring disparities in mortality rates is essential for understanding the leading causes of death in the United States from 1999 to 2017. A study by pinheiro [15] examines differences in kidney cancer death rates among various Hispanic groups in the US, such as Cuban, Puerto Rican, and Mexican. The research highlights that US-born Mexican Americans and American Indians have higher mortality rates compared to non-Hispanic Whites. The study also finds that these disparities are partly due to higher smoking and obesity rates, particularly among men. However, other factors, possibly genetic, may also contribute to these higher rates. This research underlines the need for more focused studies to understand why these groups are at greater risk. [15]

2.2 Racial/Ethnic Disparities in Drug Overdose Mortality Among US-born and Immigrant Populations

Analyzing disparities in mortality rates is essential for understanding the factors driving the leading causes of death in the United States from 1999 to 2017. These differences highlight the social, demographic, and behavioral influences on health outcomes. A study by Cano et al. [5] explores drug overdose mortality among various racial and ethnic groups in the US, comparing rates between US-born individuals and immigrants. This study is highly relevant to our research, as it underscores the impact of race, ethnicity, and immigration status on mortality risks. Cano et al. reported that US-born Black men face the highest drug overdose mortality rates, while foreign-born individuals generally exhibit lower rates across all groups. Among immigrants, White men and women have the highest overdose mortality rates. These findings demonstrate the importance of incorporating demographic factors into public health analyses, aligning with our focus on understanding mortality trends across diverse populations.

2.3 Analysis of Femicide Trends in South Africa: Intimate and Non-intimate Partner Cases (1999–2017)

Understanding the dynamics of gender-based violence is important for analyzing mortality trends, particularly those related to femicide for example a research by abrahams [1] investigates femicide (the killing of women) in South Africa, focusing on cases involving intimate partners (IPF) and non-intimate partners (NIPF) from 1999 to 2017. The study shows a decrease in all forms of femicide during this period, with a more significant decline in NIPF cases. It suggests that social and policy measures to reduce intimate partner violence (IPV) and improve social stability may have contributed to this decline. The study also emphasizes the importance of dedicated surveys in understanding gender-based violence and the need for continued efforts to prevent it. [1]

2.4 Regional Comparison of Hepatitis A Outbreaks and Vaccination Strategies in Spain (2010–2018)

Vaccination strategies are crucial in managing disease outbreaks and improving public health, which directly affects mortality trends. For instance, a study by Dominguez et al. [7] examines hepatitis A outbreak rates in different Spanish regions that adopted varying vaccination strategies. The study found that regions with universal vaccination experienced lower outbreak rates compared to those that focused on vaccinating high-risk groups. However, regions with universal vaccination also saw a higher incidence of imported outbreaks. These findings highlight the greater effectiveness of universal vaccination in reducing local outbreaks while emphasizing the importance of vaccinating high-risk groups, regardless of the approach. This research is relevant to studies of mortality trends as it demonstrates the critical role vaccination policies play in disease prevention and public health outcomes.

2.5 Impact of Environmental and Operational Factors on Activated Sludge System Microbiota: A Scientometric Analysis

Environmental factors significantly influence public health outcomes, including mortality trends. Wastewater treatment systems play a critical role in preventing waterborne diseases that can impact mortality rates. A scientometric review by Mezzalira et al. [13] investigates microbial diversity in activated sludge systems used in wastewater treatment, synthesizing 30 years of research. The study reveals that microbial communities are highly adapted to specific environmental and operational conditions, such as waste type and geographic location. Understanding this diversity is vital for improving treatment systems and reducing environmental contamination that may contribute to public health risks. These findings are particularly relevant to mortality studies, as they highlight the connection between environmental health and population-level outcomes.

2.6 Personalized Approaches to Women's Cardiovascular Health

Exploring disparities in health outcomes is a crucial aspect of analyzing the leading causes of death in the United States from 1999 to 2017, especially when considering underrepresented populations. The Research Goes Red (RGR) initiative, launched by the American Heart Association and Verily, serves as an example of efforts to address these gaps by focusing on cardiovascular disease in women [10]. This initiative is directly relevant to our research, as it demonstrates the value of targeted, population-specific studies in understanding and addressing mortality patterns. Through its registry of over 15,000 participants, RGR facilitates research on topics such as weight gain during menopause and the role of social media in increasing awareness of heart disease in women. By prioritizing the inclusion of diverse groups, the initiative highlights the need for personalized public health strategies, aligning with the goals of our study to uncover and analyze trends across different demographics.

2.7 Transcriptional Adaptations in Neural Systems Under Chronic Stress

Understanding the biological mechanisms underlying chronic health conditions is essential for identifying potential therapeutic targets for eaxample a study by basiri [2] explores how chronic conditions like obesity and drug withdrawal affect gene expression in specific brain regions. Using advanced techniques like single-cell RNA sequencing, the research identifies how different types of neurons respond to these conditions. The findings show that certain neurons are more sensitive to chronic pressures, which might contribute to the development of related health issues. Understanding these changes could lead to better treatments for conditions like obesity and addiction. [2]

2.8 Data-Driven Approaches for Longitudinal Patient Summaries

The rapid growth of digital health records has paved the way for extensive research in health informatics, particularly in the areas of data analysis and visualization. By leveraging machine learning, statistical analysis, and visual analytics, researchers aim to optimize the use of patient data to improve healthcare outcomes. One approach to this is developing interactive visualization tools that summarize a patient's medical history, emphasizing significant events based on comparisons with similar patients. These tools rely on models trained on patient populations to identify and prioritize events associated with specific conditions, which can then be used to highlight important details in a patient's medical history. [12]

2.9 3D Visualization Recommendations for Liver Disease Diagnosis and Management

Three-dimensional (3D) visualization technologies are also gaining traction in medical diagnosis and treatment, especially for liver diseases. These technologies enable clinicians to create and interpret detailed 3D models from CT scans, enhancing their ability to understand complex anatomical structures. This has proven particularly useful in liver disease management, where 3D models can assist in pre-surgical planning, virtual surgeries, and non-invasive diagnostics. Despite their potential, the lack of standardized procedures for generating and using 3D visualizations remains a challenge. [8]

2.10 Geospatial Insights for Public Health Analysis

Spatial databases and Geographic Information Systems (GIS) are critical tools in public health for managing and analyzing spatial data. These systems integrate health outcome data with geographic information to create maps and visual tools that help researchers and policymakers understand the spatial distribution of health issues. These visualizations are essential for interpreting the complex relationships between health data and geographical factors, aiding in public health decision-making. [3]

2.11 Real-Time 3D Heart Holograms for Remote Medical Diagnostics

Emerging technologies like Mixed Reality (MR) and holography are revolutionizing medical diagnostics by allowing the creation of high-definition 3D models of organs, which can be manipulated in real-time. These technologies are particularly useful in cardiology, where detailed 3D models of the heart can be generated from coronary tomography and viewed in a shared holographic environment. This innovation has the potential to improve diagnostic accuracy and enhance collaborative medical care. [4]

2.12 Global COVID-19 Surveillance Using Spatiotemporal Balloon Charts

During the COVID-19 pandemic, spatiotemporal visualization techniques were used to track the spread of the virus globally. By integrating various epidemiological indicators into balloon charts and area charts, researchers were able to visualize the pandemic's progression across different regions over time. This approach provided valuable insights into the transmissibility and severity of the virus, helping public health officials and policymakers make informed decisions. [14]. A study by pang [14] is relevant to this study on the leading causes of death in the United States from 1999 to 2017, as it mainly highlights the importance of spatiotemporal data in understanding disease patterns and trends. The ability to visualize health data dynamically helps identify regional disparities in disease outcomes, a key factor in analyzing mortality rates across different populations.

2.13 3D Visualization and Interactive Mapping of Space-Time Cancer Data

The need to integrate space-time dimensions in health data visualization is increasingly recognized, particularly in cancer research. Traditional methods often separate spatial and temporal data, limiting the ability to fully understand the interactions between geography and disease progression. New approaches aim to combine these dimensions, using 3D visualization and contextual mapping to better represent the complex factors influencing health outcomes. [11]

2.14 Advanced 3D Visualization and Printing in Radiology

Advancements in 3D visualization and 3D printing are transforming radiology by enabling the creation of detailed anatomical models that enhance the understanding of complex structures. These technologies not only aid in surgical planning but also improve patient care by providing clinicians with more accurate tools for diagnosis and treatment planning. The integration of augmented reality and virtual reality further enhances these capabilities, offering new ways to interact with and interpret medical data. [9]. This is especially relevant to this study, as it underscores how advancements in medical technology, such as 3D modeling and immersive data visualization, can enhance the precision of diagnoses and treatment planning, which in turn can influence health outcomes and mortality rates. Recognizing these technological innovations is essential for understanding how medical interventions may shape mortality trends.

2.15 Mapping Maternal Mortality and Sociodemographic Factors in Texas, 2012-2013

The relationship between social determinants and health outcomes is a growing area of research, particularly in maternal health. By mapping maternal mortality against sociodemographic indices, researchers can identify patterns and correlations that highlight the impact of social and economic factors on health. Visualizing this data at the county level provides a clearer understanding of how these determinants influence maternal health across different regions, offering valuable insights for public health interventions. [6]. This is relevant to this study, as it emphasizes the role of social and economic factors in shaping mortality trends. By understanding how these determinants affect specific populations, we can better interpret variations in mortality rates and design targeted interventions to address disparities in health outcomes.

3 DESIGN

The Five-Design-Sheet (FDS) methodology, proposed by Roberts [16], was employed as a systematic framework for generating, evaluating, and refining design concepts. This widely recognized approach supports iterative brainstorming and sketching across multiple stages, ensuring thoughtful exploration of potential solutions before implementation. By utilizing this methodology, a comprehensive range of design ideas was examined, facilitating the selection of the most effective visualization approach.

Design Process and Solutions Explored:

Sheet 1:

The initial sheet served as a brainstorming platform for generating diverse visualization ideas. These included bar charts for comparison, a 3D USA map with color-coded disease intensity (red indicating high and green indicating low), candle bar charts, heatmaps on a 3D globe, and volcanic heatmaps on a globe. These sketches provided a comprehensive range of approaches to consider.

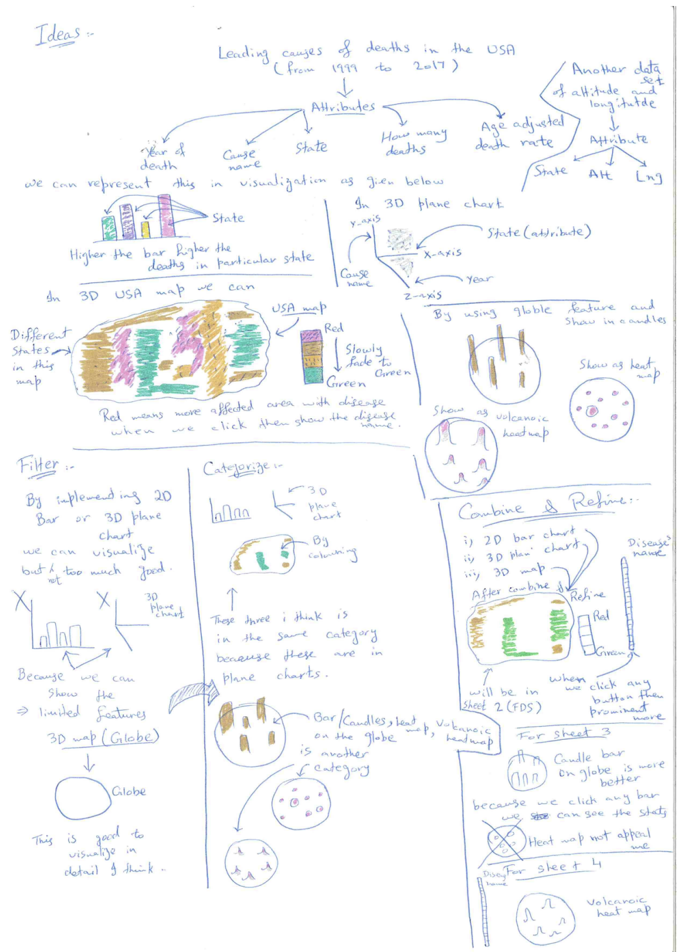

Fig. 1: FDS Sheet 1

Sheet 2:
This sheet introduced a 2D plane map of the USA, featuring interactive filters for diseases, years, and intensity levels. While this design offered clarity and simplicity, it lacked the depth required for a more comprehensive spatial and temporal analysis.

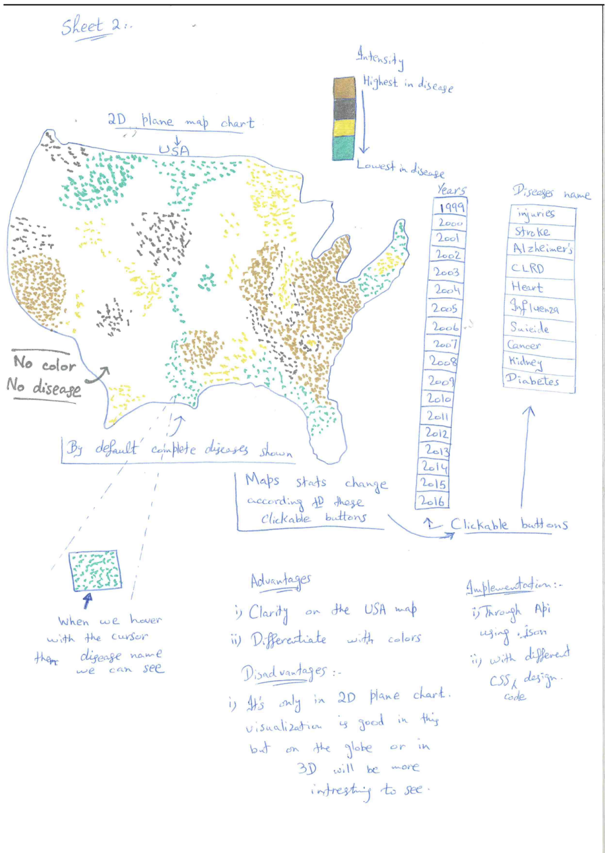

Fig. 2: FDS Sheet 2

Sheet 3:
The third sheet incorporated candle bar charts on a 3D globe, complemented by filters for diseases and years. Although the 3D perspective added dimensionality and highlighted disease intensity over time, the design was visually dense and less intuitive compared to simpler visualizations.

Sheet 4:
This sheet explored the use of a volcanic heatmap on a 3D globe, with filters for diseases and years. The heatmap effectively highlighted hotspots and trends on a global scale. However, the lack of state-specific filters limited its ability to provide detailed regional insights.

Sheet 5:
The final design combined the strengths of the previous iterations, presenting a volcanic heatmap on a 3D globe with additional filters for diseases, states, and years. This solution allowed for a comprehensive, interactive visualization of disease patterns across spatial and temporal dimensions. The finalized design was implemented using Visual Studio and JavaScript to ensure both functionality and user engagement.

Evaluation and Final Decision:

Bar charts and 2D plane maps, while easy to interpret, were insufficient for visualizing complex, multidimensional data. The 3D globe with candle bar charts added depth but was visually cluttered and harder to navigate. The volcanic heatmap on a 3D globe emerged as the most effective solution, offering an intuitive and visually compelling way to integrate spatial, temporal, and intensity-based data.

The design from Sheet 5 was finalized and implemented, demonstrating the effectiveness of the FDS methodology in developing an

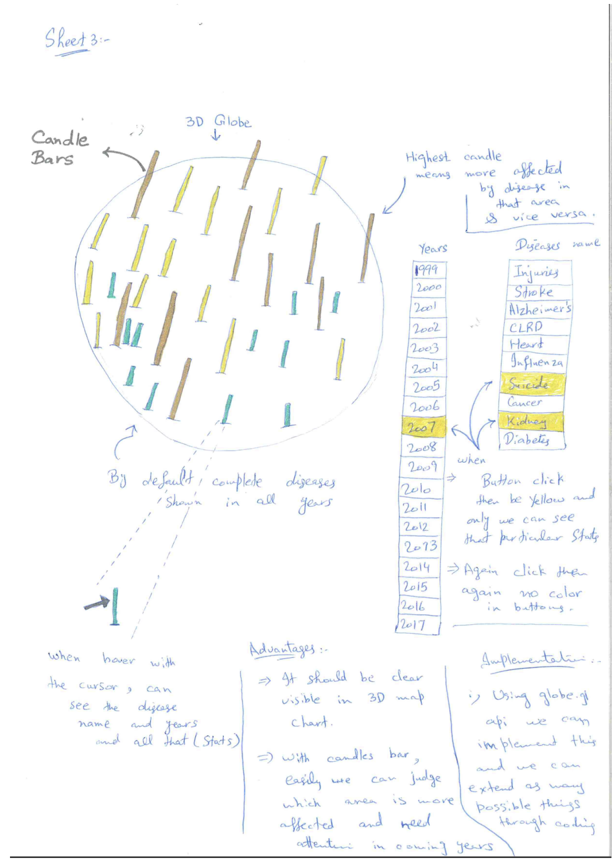

Fig. 3: FDS Sheet 3

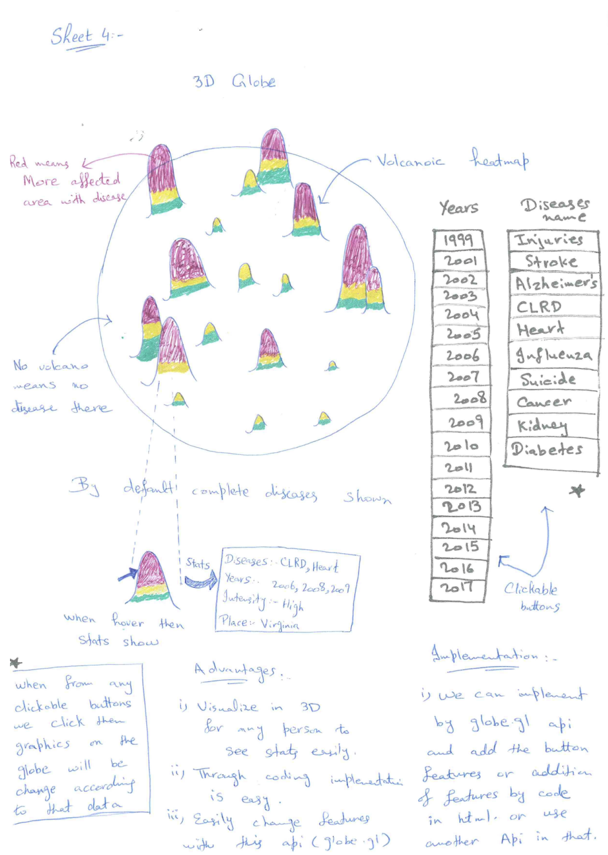

Fig. 4: FDS Sheet 4

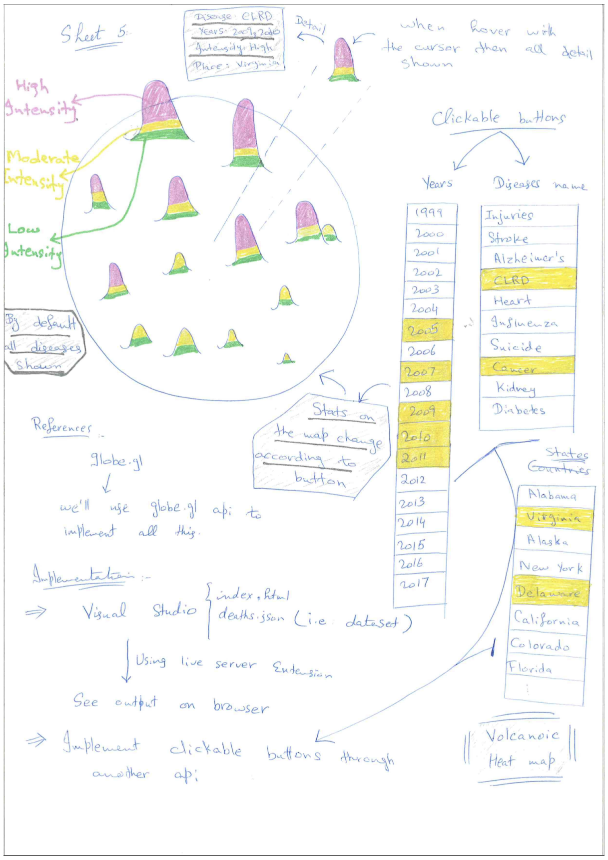

Fig. 5: FDS Sheet 5

engaging and informative visualization that met the project's objectives.

4 IMPLEMENTATION:

For this project, the dataset was obtained from the official U.S. government website, DATA.GOV, specifically the "NCHS - Leading Causes of Death: United States" dataset, accessible at https://catalog.data.gov/dataset/nchs-leading-causes-of-death-united-states.

The dataset includes six essential columns: Year, Cause Name, State, Deaths, and Age-Adjusted Death Rate. Initially available in .csv format, the dataset was converted to .json format using the VRIA platform to facilitate seamless integration. The implementation was carried out using HTML, javascript with the .json file serving as the main data source for visualization, enabling efficient representation and analysis of the information. The project was developed using Visual Studio and

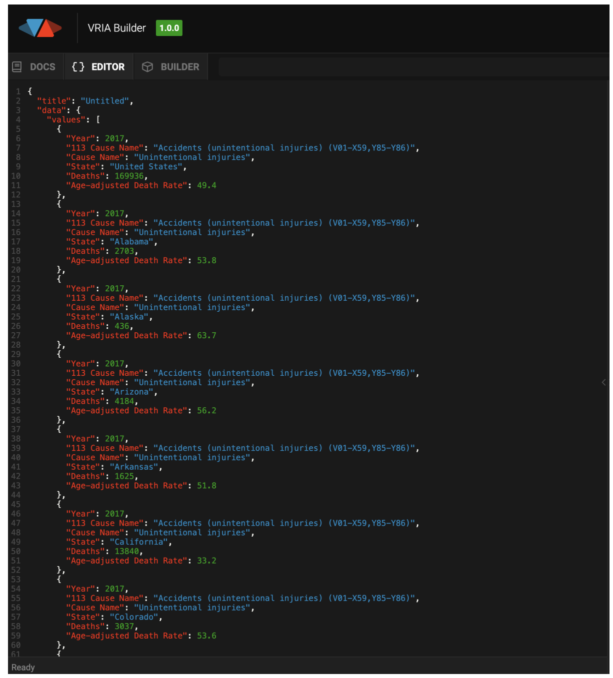

Fig. 6: extraction of .json from .csv file through vria

involved creating several key files: index.html, index.js, deaths.json, and latlngdata.json. The deaths.json file contained mortality data, while latlngdata.json stored the latitude and longitude coordinates for U.S. states. These two .json files were used independently to supply data for the visualization.

The implementation began with index.html and index.js, where the core functionality was built. The project utilized the GLOBE API from globe.gl to create an interactive 3D visualization. Features such as heatmaps and point-based visualizations were integrated to represent the data effectively. The front-end styling was achieved using CSS (Cascading Style Sheets), ensuring a polished and user-friendly interface.

The specific heatmap features implemented in the project are described below.

```
// Heatmap Layer
.heatmapsData([transformedDataset])
.heatmapPointLat('lat')
.heatmapPointLng('lng')
.heatmapPointWeight('weight') // The key to use for the heatmap weight
.heatmapTopAltitude(1)
.heatmapBandwidth(1.35)
.heatmapColorSaturation(3.2)
.onHeatmapClick(heatmap)
```

Fig. 7: Heatmap Features

Now, given below are the "point" features.

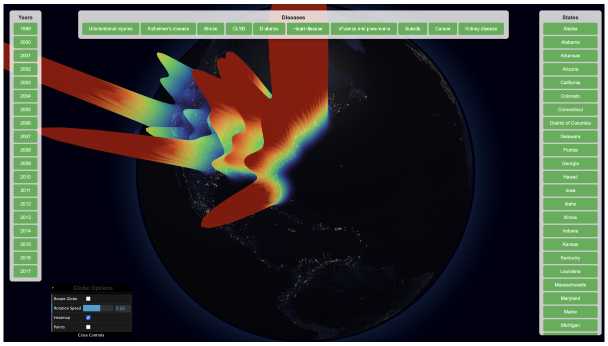

Fig. 8: Heatmap Feature Output

```
.pointsData(pointsData)
.pointLabel((d) => `<strong>${d.state}</strong><br>Deaths: ${d.weight}`)
.pointColor(() => 'rgba(255, 165, 0, 0.8)') // Orange color for points
.pointAltitude(0.1) // Altitude of the points
.pointRadius(0.2) // Radius of points
.onPointClick((d) => {
    console.log(`Clicked on point: ${d.state}`);
})
```

Fig. 9: Points Features

Fig. 10: Points Feature Output

In this project, a feature was implemented to allow users to view specific diseases, years, or states at a time by using button controls for dynamic updates. These controls enable users to filter and analyze specific regions or retrieve targeted information directly on the globe visualization.

The functionality to update the globe based on the selected filters was programmed in the index.js file, where filter logic and event handling were implemented. The corresponding frontend buttons, designed for interaction, were created in the index.html file. These buttons are visible in the application's interface, providing an intuitive way for users to apply filters and instantly observe changes on the globe.

Below is the code demonstrating how the globe is updated dynamically when filters are applied. This showcases the seamless integration of filter controls with the 3D visualization.

```
function updateGlobeData() {
    const selectedYears = Array.from(document.querySelectorAll('.year-button.active')).map(btn => +btn.innerText);
    const selectedDiseases = Array.from(document.querySelectorAll('.disease-button.active')).map(btn => btn.innerText);
    const selectedStates = Array.from(document.querySelectorAll('.state-button.active')).map(btn => btn.innerText);

    let filteredData = mergedDataset;

    if (selectedYears.length > 0) filteredData = filteredData.filter(el => selectedYears.includes(el.year));
    if (selectedDiseases.length > 0) filteredData = filteredData.filter(el => selectedDiseases.includes(el.causeName));
    if (selectedStates.length > 0) filteredData = filteredData.filter(el => selectedStates.includes(el.state));

    world.heatmapsData([filteredData]);
}

// Function to toggle button color between green and yellow and apply filters
function handleButtonClick(event) {
    const button = event.target;

    // Toggle 'active' class
    button.classList.toggle('active');

    // Update globe data based on current filters
    updateGlobeData();
}
```

Fig. 11: Update globe by using filters

Fig. 12: Update globe by using Disease filters

A new feature was added to the project by incorporating control options using Dat.GUI, positioned at the bottom left of the interface. These controls allow users to interact with the globe's rotation, offering the ability to toggle the rotation on and off. Additionally, users can adjust the rotation speed, choosing to rotate the globe either clockwise or counterclockwise.

Another useful feature included in the controls is a checkbox that enables users to hide the heatmap, allowing them to focus solely on the point layer visualization. This flexibility enhances the user experience by providing options to customize the display according to their preferences.

Below is the code implementation for these interactive controls, demonstrating how users can adjust the globe's rotation and toggle the visibility of the heatmap layer.

5 Critical Self-Evaluation

This self-evaluation section provides a critical assessment of the features developed in this project, focusing on their functionality, effectiveness, and areas for improvement. The primary tools—namely the 3D globe with a volcanic heatmap and interactive point-based filters—are evaluated based on their strengths, limitations, and possible enhancements.

I. 3D Globe Visualization: The 3D globe serves as a dynamic platform for exploring mortality data across the United States, offering interactive elements that allow users to navigate and analyze trends over time and space.

- **Strenghts:** Users can rotate, zoom, and explore data intuitively, creating an immersive experience. The ability to filter by year helps identify trends and changes over time, making it easier to spot temporal patterns. The 3D format adds depth, making the visualization more engaging and impactful compared to traditional 2D representations.

Fig. 13: Update globe by using Year filters

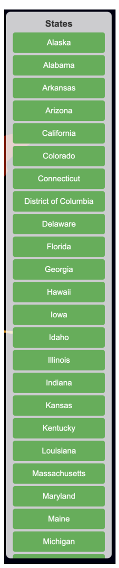

Fig. 14: Update globe by using State filters

Fig. 15: Dat GUI

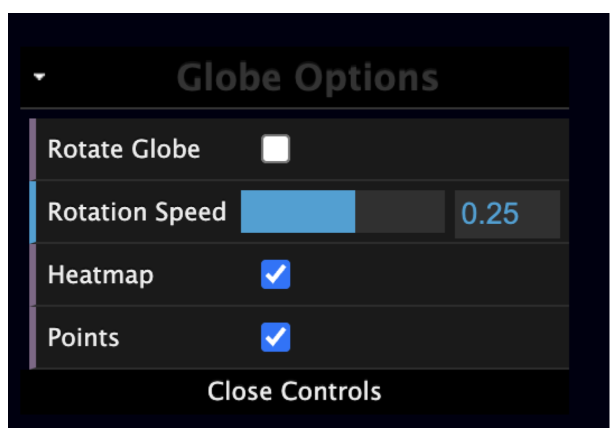

Fig. 16: Controls

- **Limitations:** Handling large datasets can slow down the visualization, especially on devices with limited processing power.
- **Suggested Improvements:** Optimizing data rendering and loading processes could improve performance and accessibility on a broader range of devices.

II.Volcanic Heatmap: The volcanic heatmap overlays the globe, using color gradients to highlight areas with higher mortality rates, effectively drawing attention to regional disparities.

- **Strenghts:** The heatmap's color intensity helps users quickly identify regions with high mortality rates. Users can apply filters based on diseases and years, allowing for focused analysis.
- **Limitations:** Subtle differences in color may be difficult to interpret, especially for users analyzing minor variations.
- **Suggested Improvements:** Enhancing color contrast and including a more detailed legend could improve clarity.

III. Interactive filters and controls: The Heatmap layer allows users to view detailed statistics by selecting specific regions, while interactive controls enable filtering by year, disease, and state.

- **Strenghts:** Hovering on data points provides detailed, region-specific information, adding depth to the analysis.
- **Limitations:** While the tool provides data points, it lacks contextual explanations that could help users interpret the findings. Button-based controls are not intuitive, impacting the user experience.
- **Suggested Improvements:** Adding pop-up descriptions with brief contextual insights would help users better understand the data. Replace buttons with a slider for smoother navigation and enhanced usability.

IV. User experience and accessibility: Overall, the project is designed to be user-friendly, but certain aspects of accessibility and device compatibility need further attention.

- **Strenghts:** The interactive design encourages user exploration and engagement with the data. Users can toggle between heatmap and point-based layers and adjust the globe's rotation, allowing for a tailored viewing experience.
- **Limitations:** The reliance on visual elements may pose challenges for users with visual impairments. Performance issues on older or less powerful devices can limit the tool's accessibility. The tool is not fully functional on mobile devices, limiting accessibility on smaller screens.
- **Suggested Improvements:** Incorporating alternative formats, such as text-based summaries or audio descriptions, could improve accessibility. Enhancing optimization to support a wider range of devices would broaden the tool's usability. Optimize the tool to support a wider range of devices, including mobile platforms.

6 Discussion:

This study aimed to identify significant patterns in mortality data through innovative visualization techniques, focusing on interactive 3D globe with a volcanic heatmap. This method was designed to uncover insights that traditional approaches often overlook, especially in analyzing trends across time and geographic regions.

In addressing the first research question, The Five Design Sheet(FDS) approach guided the exploration of various design options, including bar charts, heatmaps, and spatial visualizations. After evaluating these possibilities, interactive 2D maps and the 3D globe with a volcanic heatmap were chosen for their effectiveness in presenting complex data. The 2D maps enabled users to filter by year and disease, offering a dynamic analysis tool, while the 3D globe provided a global perspective, using color intensity to indicate high-mortality areas and making patterns easier to interpret.

For the second research question, The chosen methods proved valuable. The visualizations allowed users to explore mortality data interactively, revealing high-risk areas and disease-specific trends. The inclusion of a point layer for detailed, localized data enhanced the analysis, enabling policymakers and health officials to focus on regions requiring targeted interventions. The ability to filter data by time, states and disease further supported the identification of emerging patterns and priorities for resource allocation.

Overall, the study met its objectives by uncovering patterns that might not be visible through standard charts and tables. The 3D globe and volcanic heatmap provided a compelling and accessible way to explore mortality data, offering meaningful insights for public health strategies.

7 Conclusion:

This study demonstrated the effectiveness of advanced visualization techniques, specifically interactive 3D globe with a volcanic heatmap, in identifying patterns within mortality data. By highlighting both temporal and regional trends, these tools successfully addressed the research questions and provided valuable insights for public health decision-making.

The interactive and user-friendly design of the visualizations allowed for a more comprehensive understanding of mortality trends, facilitating data-driven strategies. This project empowered users to examine the data from multiple angles, enabling policymakers and health officials to pinpoint high-risk areas and prioritize targeted interventions to reduce preventable deaths.

Future work could enhance these visualizations by incorporating population data to adjust mortality rates, improving comparisons between regions. Additionally, integrating predictive models and real-time data could further extend the usefulness of these tools for decision-making. This study underscores the transformative potential of advanced data visualization in the analysis of complex public health data, offering a powerful tool to improve resource allocation and public health outcomes.

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